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Write Up

Introduction:

In an increasingly data-driven world, visualizations have become essential tools for making sense of complex information. They not only help interpret patterns and trends but also enable effective communication of insights to diverse audiences. However, creating impactful visualizations requires more than simply plotting data, it involves choosing the right methods, ensuring accessibility, and understanding how different visualizations align with specific datasets.

Our project dives into key questions about the role and design of data visualizations:

- Why are visualizations critical for representing data effectively?
- What types of visualizations are available, and how can they be used?
- How can visualizations be made accessible to a wider audience?
- How do similar visualizations differ in their application depending on the data?
- What are the consequences of using an inappropriate visualization for a dataset?

By addressing these questions, this project aims to provide practical guidance for creating clear, accurate, and accessible visualizations, empowering users to present their data in meaningful and impactful ways.

Approach:

Our project is divided into three main sections: Different Visualizations, Accessibility, and Comparison Between Visualizations. In the Different Visualizations section, we explained six types of visualizations: bar graphs, scatter plots, line graphs, box plots, heatmaps, and forced connection plots, going over their benefits, typical use case. The Accessibility section focused on addressing a common accessibility issue in data visualization and proposing a practical solution to ensure inclusivity for all users. Finally, the Comparison Between Visualizations section compared three pairs of visualizations with similar functionalities or visual characteristics, emphasizing how differences in design or application can influence their effectiveness depending on the dataset. Together, these sections provide a comprehensive framework for understanding, designing, and evaluating data visualizations.

Code:

Bar graph:

```
df = pd.read_csv('../input/palmer-archipelago-antarctica-penguin-data/penguins_size.csv')

df.head()

df['species'].value_counts().iplot(kind='bar')
```

Scatter Plot:

```
selected_columns2 <- data_200m[, c("Athlete", "Wind","Time")]


UofA2$School = 'UofA2'

ASU2$School = 'ASU2'

NAU2$School = 'NAU2'

UW2$School = 'UW2'

Cal2$School = 'Cal2'

CU2$School = 'CU2'

WSU2$School = 'WSU2'

SU2$School = 'SU2'

USC2$School = 'USC2'


Merged_200m_data <- bind_rows(UofA2,ASU2,NAU2,UW2,Cal2,CU2,WSU2,SU2,USC2)


# correlation coefficient

correlation <- cor(Merged_200m_data$Wind, Merged_200m_data$Time, use = "complete.obs")


# Load necessary library

library(ggplot2)


# Create a scatter plot with regression line

ggplot(data_200m, aes(x = Wind, y = Time)) +

  geom_point(size = 3, alpha = 0.7, color = "blue") +  # Scatter plot with points

  geom_smooth(method = "lm", se = FALSE, color = "red") +  # Linear regression line

  scale_y_reverse() +  # Reverse the y-axis so higher times are at the top
```

```
  labs(
    title = "Scatter Plot: Wind Speed vs 200m Time",
    x = "Wind Speed (m/s)",
    y = "200m Time (seconds)"
  ) +
  theme_minimal() +  # Clean theme
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    axis.title = element_text(size = 14)
  )
```

Line Graph:

Load necessary libraries

```
library
library(dplyr)
```

Load the data from CSV file

```
data <- read.csv("C:/Users/aasth/OneDrive/data viz/project-2-infocrew/data/world-education-data.csv")

pakistan_data <- data %>% filter(country_code == "PAK")
```

Inspect the data (optional)

```
head(pakistan_data)
```

Create the plot for Pakistan with specific colors for each line

```r
ggplot(pakistan_data, aes(x = year)) +
  geom_line(aes(y = gov_exp_pct_gdp, color = 'Govt Expenditure (% of GDP)'), size = 1) +
  geom_point(aes(y = gov_exp_pct_gdp, color = 'Govt Expenditure (% of GDP)'), size = 2) +
  geom_line(aes(y = lit_rate_adult_pct, color = 'Literacy Rate for Adults (%)'), size = 1) +
  geom_point(aes(y = lit_rate_adult_pct, color = 'Literacy Rate for Adults (%)'), size = 2) +
  geom_line(aes(y = pri_comp_rate_pct, color = 'Primary Completion Rate (%)'), size = 1) +
  geom_point(aes(y = pri_comp_rate_pct, color = 'Primary Completion Rate (%)'), size = 2) +
  labs(title = "Trends of Educational Indicators in Pakistan (1999-2004)",
       x = "Year",
       y = "Percentage",
       color = "Indicators") +
  scale_color_manual(values = c('Govt Expenditure (% of GDP)' = 'blue',
                                'Literacy Rate for Adults (%)' = 'green',
                                'Primary Completion Rate (%)' = 'red')) +
  theme_minimal() +
  theme(legend.position = "top")
```

Box Plot:

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from scipy.stats import norm
```

```
from scipy import stats

import warnings

warnings.filterwarnings('ignore')

%matplotlib inline

Input data files are available in the "../input/" directory on kaggle.

from subprocess import check_output

print(check_output(["ls", "../input"]).decode("utf8"))

var = 'Country_code'

data_plt = pd.concat([data_scale['mpg'], data_scale[var]], axis=1)

f, ax = plt.subplots(figsize=(8, 6))

fig = sns.boxplot(x=var, y="mpg", data=data_plt)

fig.axis(ymin=0, ymax=1)

plt.axhline(data_scale.mpg.mean(),color='r',linestyle='dashed',linewidth=2)
```

Heatmap:

```
# Load necessary libraries

library(ggplot2)

library(reshape2)

library(RColorBrewer)


# Load the data from CSV file

data <- read.csv("C:/Users/aasth/OneDrive/data viz/project-2-infocrew/data/world-education-data.csv")


# Select relevant columns for correlation

data_corr <- data[, c("gov_exp_pct_gdp", "lit_rate_adult_pct", "pri_comp_rate_pct",

                     "pupil_teacher_primary", "pupil_teacher_secondary",
```

```
                    "school_enrol_primary_pct", "school_enrol_secondary_pct",
                    "school_enrol_tertiary_pct")]

# Calculate the correlation matrix
corr_matrix <- cor(data_corr, use = "complete.obs")

# Reshape the correlation matrix for ggplot
corr_melt <- melt(corr_matrix)

# Create the correlation heatmap
ggplot(corr_melt, aes(Var1, Var2, fill = value)) +
  geom_tile() +
  scale_fill_gradientn(colors = brewer.pal(11, "Spectral")) +  # Custom color scheme
  labs(title = "Correlation Heatmap of Educational Indicators",
       x = "",
       y = "",
       fill = "Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        axis.text.y = element_text(angle = 0, hjust = 1)) +
  coord_fixed()
```

Forced Connected Plot:

Load libraries

library(networkD3)

library(webshot)

library(htmlwidgets)

library(htmltools)

Read data

data <- read.csv("C:/Users/aasth/OneDrive/data viz/project-2-infocrew/data/imdb_edgelist.csv",

nrows = 500, header = TRUE, stringsAsFactors = FALSE)

Check if data is read correctly

if (is.null(data) nrow(data) == 0) {

stop("Failed to read data.")

}

Unique nodes

nodes <- unique(c(data$From, data$To))

nodes_df <- data.frame(name = nodes, group = 1) # Add a 'group' column for compatibility

Map node names to IDs

data$FromID <- match(data$From, nodes) - 1

data$ToID <- match(data$To, nodes) - 1

Check if IDs are mapped correctly

if (any(is.na(data$FromID)) any(is.na(data$ToID))) {

stop("Failed to map node names to IDs.")

}

Create force network graph

graph <- forceNetwork(

Links = data.frame(source = data$FromID, target = data$ToID, value = data$Strength),

Nodes = nodes_df,

Source = "source",

Target = "target",

Value = "value",

NodeID = "name",

Group = "group",

fontSize = 16,   # Increase font size

fontFamily = "Arial",   # Use a bold-compatible font like Arial

opacity = 0.9,   # Transparency for nodes

zoom = TRUE,   # Enable zoom

linkDistance = 100,   # Distance between nodes

opacityNoHover = 0.1   # Highlight connections on hover

)

Save the force network graph as an HTML file

html_file <- "force_graph.html"

saveNetwork(graph, file = html_file, selfcontained = TRUE)

Convert the saved HTML to PNG using webshot

webshot::webshot(html_file, file = "force_graph.png", vwidth = 1200, vheight = 800)

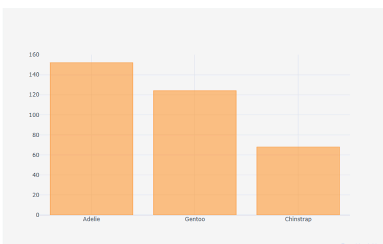
Accessibility Section: (All Plots)

```
{r}

library(palmerpenguins)

library(dplyr)

library(ggplot2)

{r}

head(penguins)

theme_set(theme_minimal())

{r}

ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g)) +

  geom_point(aes(color = species),

             size = 2) +

  scale_color_manual(values = c("#f27874","#00bc3e","#699afb"))

{r}

ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g)) +

  geom_point(aes(color = species,

                 shape = species),

             size = 2) +

  scale_color_manual(values = c("#f27874","#00bc3e","#699afb"))

{r}

ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g)) +

  geom_point(aes(color = species),

             size = 2) +

  scale_color_manual(values = c("#340042","#1f8079","#fce41e"))

{r}
```

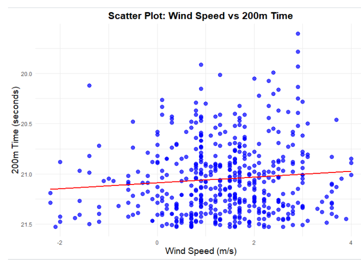
```
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g)) +
  geom_point(aes(color = species,
                 shape = species),
             size = 2) +
  scale_color_manual(values = c("#340042","#1f8079","#fce41e"))
```

Visualizations

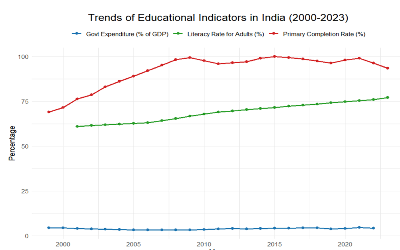
Different Visualizations Types:



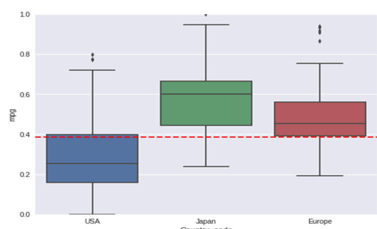
(bar)



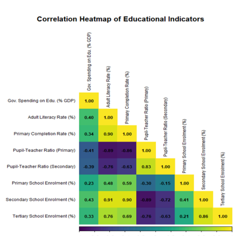
(scatter)



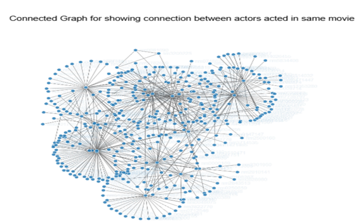
(line)



(box)

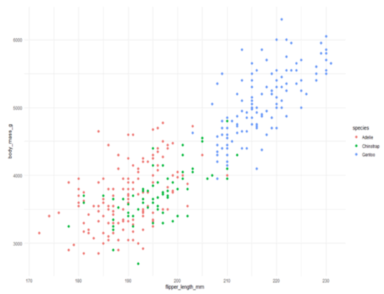


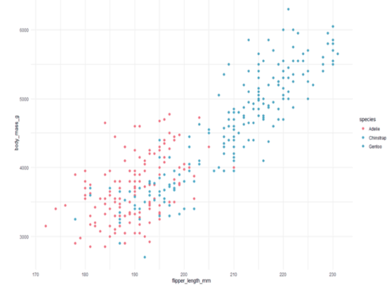
(heat)

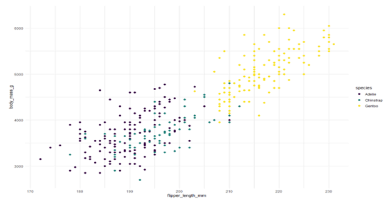


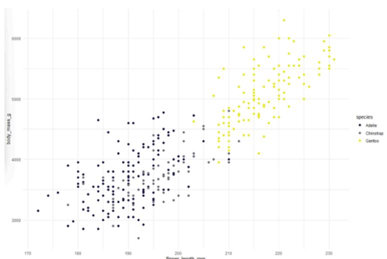
(Network plot)

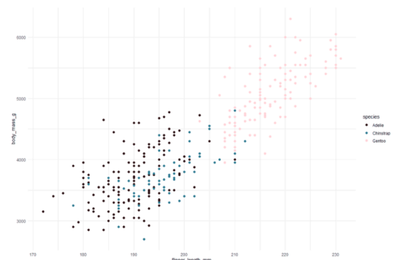
Accessibility Visualizations:

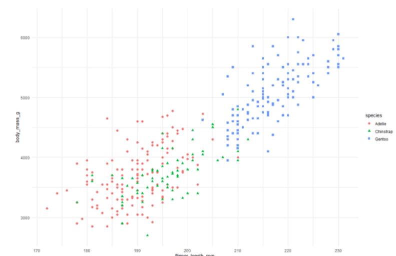




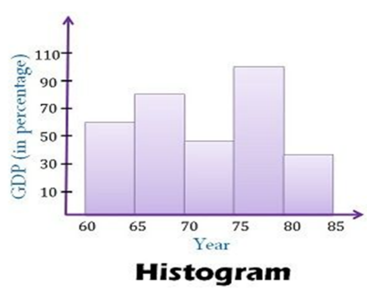


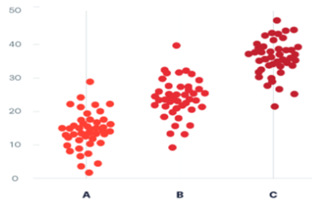




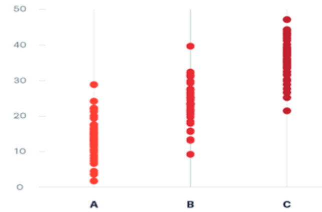


Comparison visualizations:

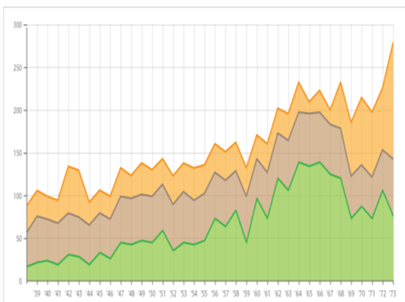




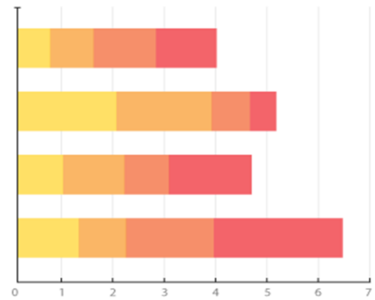
(Jitter)



(Strip)



(Stacked Area)



(Stacked Bar)

Discussion:

In our project, we aimed to provide the audience with key considerations to keep in mind before starting their visualizations. We began by introducing a selection of versatile visualization types, showcasing their benefits and use cases to help the audience choose the most suitable options for their data. Next, we addressed the accessibility aspects of visualizations, emphasizing the importance of inclusivity and usability. Finally, we explored how the context both in terms of what the visualization is expected to convey and the type of data being visualized can significantly impact the choice of visualization. To illustrate this, we provided three examples where these contextual factors influenced the effectiveness of the selected visualizations.